

LPP 3CFU – Domande d'esame A.A. 25/26
Parte 1

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November 25, 2025

CT0050 First categories

Exercise 1. Prove that any pre-order is a category.

Hint 1.

- recall def. of pre-order
- show that reflexivity implies the existence of identity using transitivity
- show that transitivity is arrow composition $\square$

Exercise 2. Write an example of pre-order and the corresponding commuting diagram to show that the pre-order is a category. $\square$

Exercise 3. Prove that any partial-order is a category. $\square$

Exercise 4. Write an example of partial-order and the corresponding commuting diagram to show that the pre-order is a category. Highlight a possible graph that cannot be a sub-graph of the commuting diagram corresponding to the given example. $\square$

Exercise 5. Prove that any total-order is a category. $\square$

Exercise 6. Write an example of total-order and the corresponding commuting diagram to show that the pre-order is a category. Highlight a possible graph that cannot be a sub-graph of the commuting diagram corresponding to the given example. $\square$

Exercise 7. Prove that at most one arrow exists between two objects of the commuting diagram corresponding to a pre-order. $\square$

Exercise 8. Prove that at most one arrow exists between two objects of the commuting diagram corresponding to a partial-order. $\square$

Exercise 9. Prove that exactly one arrow exists between two objects of the commuting diagram corresponding to a total-order. $\square$

Exercise 10. Justify why we cannot say that every oriented graph is a category. $\square$

CT0100 Categorical monoids

Exercise 11. Let $\mathbb{B} = \{T, F\}$.

Write the Categorical monoid of the monoid $(\mathbb{B}, \wedge, T)$ $\square$

Exercise 12. Let $\mathbb{B} = \{T, F\}$.

Write the Categorical monoid of the monoid $(\mathbb{B}, \vee, F)$ □

Exercise 13. Let $\mathbb{B} = \{T, F\}$.

Write the Categorical monoid of the monoid $(\mathbb{B}, \oplus, F)$ (“exclusive or”) □

Exercise 14. Write the Categorical monoid of the monoid $(\Sigma^*, \cdot, \epsilon)$ (strings) □

Exercise 15. Write the Categorical monoid of the monoid $(\{[a_1, \dots, a_n] \mid n \in \mathbb{N}\}, ++, [\])$ (lists) □

CT0150 Category Set and “beyond”

Exercise 16. Recall the definition of *monomorphism* and how it is obtained as a generalization of the notion: “injective function” between two sets. □

Exercise 17. Recall the definition of *epimorphism* and how it is obtained as a generalization of the notion: “surjective function” between two sets. □

Exercise 18. Recall the definition of “Terminal object in a generic category” and how it is obtained as a generalization of the notion: “singleton set”. □

Exercise 19. Prove that the terminal object of a category $\mathcal{C}$ is unique up to isomorphism. □

Exercise 20. Let $\{a, b, c\}$ be a set with three elements. Is it terminal in the category **Set**? In case of a positive answer, justify the answer by means of an example. Analogously, in case of a negative answer, justify the answer by means of an example. □

Exercise 21. Recall the definition of “Product between a and b ” as given in Category Theory. Then, justify why $(a, \lambda x \rightarrow x, \lambda x \rightarrow y : b)$ cannot be taken as Product between a and b . □

Exercise 22. Recall the definition of “Product between a and b ” as given in Category Theory. Then, justify why $((a, a, b), \lambda (x, _, _) \rightarrow x, \lambda (_, _, z) \rightarrow z)$ cannot be taken as Product between a and b . □

Exercise 23. Let us assume that both P and Q are the Product of a and b in a Category. Prove that P and Q are isomorphic. □

Exercise 24. Is the following triple:

```
((Bool, Int, Bool)
, \ (x, y, z) -> z
, \ (x, y, z) -> y )
```

a product between the Haskell types `Bool` and `Int`?

Exercise 25. Is the following triple:

```
((Bool, Int, Bool)
, \ (x, y, z) -> x
, \ (x, y, z) -> y )
```

a product between the Haskell types `Bool` and `Int`?

Exercise 26. Is the following triple:

```
((Bool, Int)
, \ (x, y) -> x
, \ (x, y) -> y + 1)
```

a product between the Haskell types `Bool` and `Int`?

Exercise 27. Is the following triple:

```
((Bool, Int)
, \ (x, y) -> not x
, \ (x, y) -> y)
```

a product between the Haskell types `Bool` and `Int`?

CT0200 Universal constructions

Exercise 28. Is the following triple:

```
(Either (Either Bool Int) Int
, \ x::Int -> Left (Right x)
, \ z::Bool -> Left (Left z) )
```

a Co-Product between `Int` and `Bool`?

□

Exercise 29. Is the following triple:

```
(Either Int Bool
, \ x::Int -> Left (x + 1)
, \ y::Bool -> Right y )
```

a Co-Product between `Int` and `Bool`? ☐

Exercise 30. Is the following triple:

```
(Either Int Bool
, \x::Int  -> Left x
, \y::Bool -> Right (not y)      )
```

a Co-Product between `Int` and `Bool`? ☐

Exercise 31. Is the following triple:

```
(Int
, \x::Int  -> x
, \y::Bool -> if y then 0 else 1      )
```

a Co-Product between `Int` and `Bool`? ☐

Exercise 32. Is the following triple:

```
(Int
, \x::Int  -> if x<0 then x else x+2
, \y::Bool -> if y then 0 else 1      )
```

a Co-Product between `Int` and `Bool`? ☐

Exercise 33. Justify why “`Either a Void`” and `a` are isomorphic. ☐

Exercise 34. Justify why “`(a, Either b c)`” and “`Either (a, b) (a, c)`” are isomorphic. ☐

Exercise 35. Are “`Either a a`” and “`(Either () (), a)`” isomorphic? ☐